

EDUCATION

- **Hong Kong University of Science and Technology (HKUST)** Hong Kong, China
Ph.D. student in Computer Science and Engineering. Supervisor: Prof. Dan Xu Feb. 2022 – Present
- **University of Science and Technology of China (USTC)** Hefei, China
Bachelor of Science in Automation Sep. 2015 – June 2019

EXPERIENCE

- **Tencent Inc.** Shenzhen, China
Research Intern May 2025 – Present
 - **Multimodal 3D Understanding:** Developed multimodal models for 3D spatial reasoning and geometric perception, including first-author N3D-VLM for native 3D grounding and spatial reasoning. Designed the N3D-VLM data construction pipeline, the model, and the evaluation. Also contributed to DepthVLM and Stream3D-VLM for metric depth estimation and online 3D understanding.
- **Shanghai AI Lab** Shanghai, China
Research Intern Feb. 2022 – Feb. 2023
 - **3D/4D Datasets & Benchmarks:** Co-developed OmniObject3D, one of the most popular 3D object datasets to date. Responsible for video data collection, processing, and setting up part of the benchmarks. Also co-developed RenderMe-360, a large-scale 4D head-avatar dataset and benchmark.
- **SenseTime Inc.** Beijing, China
Research Intern $\Rightarrow$ Computer Vision Researcher Sep. 2019 – Feb. 2022
 - **Text/Audio-Driven Talking Heads:** Developed and deployed text- and audio-driven 2D, 3D, and cartoon head generation algorithms across multiple products, generating more than RMB 10 million in revenue for the company. The work resulted in three patents and a first-author book chapter.

SELECTED PUBLICATIONS

3D VLMs: Spatial Reasoning & Geometric Perception

- **Yuxin Wang**, Lei Ke, Boqiang Zhang, Tianyuan Qu, Hanxun Yu, Zhenpeng Huang, Meng Yu, Dan Xu, Dong Yu. “N3D-VLM: Native 3D Grounding Enables Accurate Spatial Reasoning in Vision-Language Models.” **arXiv 2025**.
- Hanxun Yu*, Xuan Qu*, **Yuxin Wang**, Jianke Zhu, Lei Ke. “Unlocking Dense Metric Depth Estimation in VLMs.” **arXiv 2026**.
- Hanxun Yu*, Xuan Qu*, Lei Ke, Boqiang Zhang, **Yuxin Wang**, Jianke Zhu, Dong Yu. “Stream3D-VLM: Online 3D Spatial Understanding with Incremental Geometry Priors.” **ECCV 2026**.

3D Modeling & Manipulation

- **Yuxin Wang**, Qianyi Wu, and Dan Xu. “F3D-Gaus: Feed-forward 3D-aware Generation on ImageNet with Cycle-Aggregative Gaussian Splatting.” **arXiv 2025**.
- **Yuxin Wang**, Qianyi Wu, Guofeng Zhang, and Dan Xu. “GScream: Learning 3D Geometry and Feature Consistent Gaussian Splatting for Object Removal.” **ECCV 2024**.
- **Yuxin Wang**, Wayne Wu, and Dan Xu. “UDC-NeRF: Learning Unified Decompositional and Compositional NeRF for Editable Novel View Synthesis.” **ICCV 2023**.
- Zhenxing Mi, **Yuxin Wang**, Dan Xu. “One4D: Unified 4D Generation and Reconstruction via Decoupled LoRA Control.” **ECCV 2026**.
- Shaocong Dong, Lihe Ding, Xiao Chen, Yaokun Li, **Yuxin Wang**, Yucheng Wang, Qi Wang, Jaehyeok Kim, Chenjian Gao, Zhanpeng Huang, Zibin Wang, Tianfan Xue, Dan Xu. “From One to More: Contextual Part Latents for 3D Generation.” **ICCV 2025**.

3D/4D Datasets & Benchmarks

- Tong Wu, Jiarui Zhang, Xiao Fu, **Yuxin Wang**, Jiawei Ren, Liang Pan, Wayne Wu, Lei Yang, Jiaqi Wang, Chen Qian, Dahua Lin, and Ziwei Liu. “OmniObject3D: Large-Vocabulary 3D Object Dataset for Realistic Perception, Reconstruction and Generation.” **CVPR 2023 (Award Candidate: 12/9155)**.
- Dongwei Pan, Long Zhuo*, Jingtian Piao*, Huiwen Luo*, Wei Cheng*, **Yuxin Wang***, Siming Fan, Shengqi Liu, Lei Yang, Bo Dai, Ziwei Liu, Chen Change Loy, Chen Qian, Wayne Wu, Dahua Lin, and Kwan-Yee Lin. “RenderMe-360: A Large Digital Asset Library and Benchmarks Towards High-fidelity Head Avatars.” **NeurIPS 2023 (Joint Second Author)**.

Technical Report

- Tencent HY Frontier Team (including **Yuxin Wang**). “UI-Mate: Advancing Open-Weight Foundation GUI Agents with In-Context Demonstrations.” **arXiv 2026**.

Book Chapter

- **Yuxin Wang**, Linsen Song, Wayne Wu, Chen Qian, Ran He, and Chen Change Loy. “Talking Faces: Audio-to-Video Face Generation.” *Handbook of Digital Face Manipulation and Detection*. **Springer 2022**.

SKILLS and SERVICES

- **Programming:** Python, C/C++, Matlab, LaTeX.
- **Languages:** Mandarin Chinese, English.
- **Conference Reviewer:** CVPR, ICCV, ICLR, NeurIPS, AAAI, and ACM MM.
- **Teaching Assistant:** Fundamentals of Artificial Intelligence; Introduction to Object-oriented Programming; Deep Perception, Localization, and Planning for Autonomous Vehicles.